



PROGRAM STUDI INFORMATIKA FAKULTAS TEKNIK UNIVERSITAS SILIWANGI

UJIAN TENGAH SEMESTER (UTS) TAHUN AKADEMIK 2026 GANJIL

Mata Kuliah : Malware Analysis
Jumlah Sks : 3 Sks
Metoda Tes : Take Home / Projects Base
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DAFTAR SOAL PROJECT-BASED — MALWARE ANALYSIS

Struktur

- 10 Proyek Individu (10 mahasiswa acak)
 - 10 Proyek Pasangan (20 mahasiswa → 10 kelompok)
 - 5 Proyek Kelompok 4 orang (siswa 20 mahasiswa → 5 kelompok)
- Total: 25 Projects (cukup untuk 40 mahasiswa)

A. PROYEK INDIVIDU (10 Topik)

Masing-masing mahasiswa mengerjakan secara mandiri.

Proyek 1 — Static Analysis Dasar pada Sample Trojan

- Lakukan *static analysis* menggunakan **strings**, **PEiD**, **Detect It Easy**, dan **PEStudio**.
- Identifikasi: compiler, packing, imports, suspicious string.
- **Output:** laporan PDF + screenshot.

Static Analysis Dasar

Malware: Lokibot (clean sample for static analysis)

Link (safe sandbox view):

🔗 <https://app.any.run/malware-trends/lokibot>

(Mahasiswa hanya melihat static info, tidak mengunduh .exe)

Proyek 2 — Dynamic Behavior Analysis (Sandboxing)

- Gunakan **ANY.RUN** / **Cuckoo Sandbox**.
- Analisis: file modification, registry modification, network traffic.
- **Output:** event summary + behavior tree.

Dynamic Analysis / Sandbox

Malware: AgentTesla

Link (safe):

🔗 <https://app.any.run/malware-trends/agenttesla>



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Proyek 3 — Network Traffic Analysis dari Malware Beaconing

- Analisis file PCAP dari malware beaconing (disediakan dosen).
- Gunakan **Wireshark**.
- Identifikasi: C2 domain, port, payload.
- **Output:** flow diagram komunikasi malware.

Network Traffic Analysis (PCAP)

Malware: Trickbot PCAP

Link:

🔗 <https://www.malware-traffic-analysis.net/2020/02/04/index.html>
(*File hanya berupa PCAP, bukan malware executable → aman *)

Proyek 4 — Analisis YARA Rules

- Buat dan uji minimal **3 custom YARA rules** untuk mendeteksi sampel malware tertentu.
- **Output:** rule file + hasil pengujian.

YARA Rules

Sample: Emotet PE sample (view only)

🔗 <https://app.any.run/malware-trends/emotet>

Atau gunakan **test malware:**

🔗 <https://github.com/ytisf/theZoo>

Proyek 5 — Reverse Engineering Dasar

- Gunakan **Ghidra / IDA Free**.
- Temukan fungsi mencurigakan, API call, dan definisi string.
- **Output:** ringkasan fungsi penting + call graph.

Reverse Engineering Dasar

Sample: Calculator malware (benign educational sample)

🔗 https://github.com/hasherezade/malware_training_vol1

Sangat aman untuk RE, dibuat khusus untuk pendidikan.

Proyek 6 — Shellcode Examination

- Analisis potongan shellcode (hex dump disediakan).
- Identifikasi: fungsi API, behavior, decoding routine.
- **Output:** penjelasan langkah-langkah eksekusi shellcode.

Shellcode Examination

Sample: Windows shellcode (benign, educational)

🔗 <https://github.com/odzhan/shellcode>



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Proyek 7 — Malware Persistence Mechanism

- Teliti 1 persistence technique (Registry Run Keys / Task Scheduler / DLL Hijacking).
- Berikan contoh PoC kecil (non-malicious).
- **Output:** mini report + demo.

Persistence Technique Non-malicious PoC

🔗 <https://github.com/SigmaHQ/sigma/tree/master/rules/windows/persistence>

Proyek 8 — Memory Forensics

- Gunakan **Volatility** untuk file memdump yang disediakan.
- Temukan: proses mencurigakan, open handles, network connections.
- **Output:** tabel temuan + kesimpulan.

Memory Forensics

Memory Dump: REvil ransomware (safe dump)

🔗 <https://www.malware-traffic-analysis.net/2021/07/06/index.html>
Volatility-compatible memory dump.

Proyek 9 — Phishing Payload Analysis

- Analisis file dokumen Office berisi macro.
- Gunakan **OLETools** / **Oledump**.
- **Output:** deobfuscated macro + alur eksekusi.

Phishing Macro Analysis

Sample: Malicious Word doc with macro (sanitized)

🔗 <https://github.com/DissectMalware/malicious-macros>

Proyek 10 — Ransomware Encryption Flow

- Analisis pseudocode ransomware sederhana (diberikan).
- Identifikasi algoritma enkripsi, key generation, file rename process.
- **Output:** diagram alur enkripsi.

Ransomware Encryption Flow

Educational Ransomware Simulation (SAFE)

🔗 <https://github.com/joeymalvinni/Educational-Ransomware>



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B. PROYEK PASANGAN (20 Mahasiswa → 10 Tim)

Setiap tim terdiri dari 2 orang.

Proyek 11 — Analisis Malware Packed vs Unpacked

- Satu orang melakukan **static analysis**,
- Satu orang melakukan **dynamic analysis**,
- Bandingkan hasil packed vs unpacked.
- Tools: UPX, PESTudio, Process Monitor.

Packed vs Unpacked Malware

Sample: UPX-packed benign test program

🔗 <https://github.com/upx/upx>

Tambahan packed malware sample (safe signature only):

🔗 <https://github.com/ytisf/theZoo>

Proyek 12 — Deep Dive into RAT (Remote Access Trojan)

- Menganalisis RAT open-source (njRAT / Quasar RAT).
- Fokus: communication protocol, modules, persistence.
- **Output:** laporan + diagram struktur RAT.

Remote Access Trojan (RAT)

Open-source RAT (legal):

- Quasar RAT → <https://github.com/quasar/Quasar>
- AsyncRAT → <https://github.com/NYAN-x-CAT/AsyncRAT-C-Sharp>

(Tidak berisi malicious payload, digunakan untuk riset)

Proyek 13 — Malware Classification Using ML

- Dataset: EMBER atau VirusShare labels.
- Lakukan ekstraksi fitur + classification menggunakan SVM / Random Forest.
- **Output:** Python notebook + confusion matrix.

ML Classification

Dataset: EMBER

🔗 <https://github.com/elastic/ember>

Proyek 14 — Detection Evasion Techniques

- Pelajari 3 teknik evasion: obfuscation, anti-debug, anti-VM.
- Uji menggunakan contoh kode sederhana.
- **Output:** demonstrasi + laporan.

Evasion Techniques



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Sample: Anti-debug PoC (safe)

 <https://github.com/CheckPointSW/Evasions>

Proyek 15 — Incident Response Simulation: Malware Infection

- Tim membuat laporan IR dari skenario kasus.
- Analisis log, PCAP, registry artifacts.
- **Output:** IR Report (MITRE ATT&CK-based).

Incident Response Simulation

Case: Emotet infection logs

 <https://www.malware-traffic-analysis.net/2020/01/13/index.html>

Proyek 16 — Comparative Study: Keylogger Techniques

- Analisis 2 metode keylogging (API hooking vs polling).
- Buat PoC tiny program (non-malicious).
- **Output:** perbandingan teknis.

Keylogger Techniques

Educational Keylogger (non-malicious PoC)

 <https://github.com/GalaxySoftwareStudio/Educational-Keylogger>

Proyek 17 — DNS Tunneling dari Malware

- Analisis PCAP malware yang menggunakan DNS tunneling.
- Temukan encoding method & data exfiltration.
- **Output:** decoding script + report.

DNS Tunneling

PCAP Malware: Iodine / DNSCat2

 <https://www.malware-traffic-analysis.net>

Proyek 18 — Macro Malware Reconstruction

- Reverse macro malware.
- Rekonstruksi timeline serangan (macro → script → payload).
- **Output:** visual attack chain (kill-chain).

Macro Malware Reconstruction

Sample: Excel dropper macro (safe repo)

 <https://github.com/DissectMalware/malicious-macros>

Proyek 19 — Botnet Activity Mapping

- Berdasarkan traffic log, identifikasi botnet's:
 - C2 server
 - heartbeat intervals
 - payload update pattern
- **Output:** botnet lifecycle mapping.



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Botnet Activity Mapping

PCAP: Trickbot / Emotet

 <https://www.malware-traffic-analysis.net/>

Proyek 20 — Android Malware Analysis

- Gunakan APKTool, jadx.
- Analisis permission abuse + malicious activity.
- **Output:** tree of malicious components.

Android Malware Analysis

Sample: APK malware dataset

 <https://github.com/ashishb/android-malware>

C. PROYEK KELOMPOK (20 Mahasiswa → 5 Kelompok @ 4 Orang)

Cocok sebagai Final Project.

Proyek 21 — Build Your Own Malware Sandbox

- Kelompok membangun mini sandbox:
 - file monitor
 - registry monitor
 - process monitor
- Tools: Python + FMonitor Lib.
- **Output:** prototype + documentation.

Build Your Own Sandbox

Tidak memerlukan malware nyata.

Gunakan executable benign test program.

 https://github.com/hasherezade/malware_training_vol1

Proyek 22 — Full Reverse Engineering of Unknown Malware Sample

- Lakukan full analysis mulai dari:
 - static
 - dynamic
 - network
 - reverse engineering
- **Output:** Malware Analysis Report (standar industri).

Full Reverse Engineering

Sampel rekomendasi (TheZoo):

- NjRAT
- Zeus
- CryptoLocker (encrypted sample)

 <https://github.com/ytisf/theZoo>

(perlu password “infected”, aman dari eksekusi accidental)

Proyek 23 — MITRE ATT&CK Mapping untuk Sebuah Malware Keluarga

- Pilih 1 malware family: Emotet / Trickbot / AgentTesla.
- Mapping seluruh behavior ke matrix ATT&CK.



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- **Output:** matrix + report + video presentasi.

MITRE ATT&CK Mapping

Malware dataset:

- Emotet
- Trickbot
- Qakbot

🔗 <https://malpedia.caad.fkie.fraunhofer.de/>

Proyek 24 — System Call Tracing & Anomaly Detection

- Buat sistem kecil pendeteksi kelainan syscall malware.
- Gunakan strace / Sysmon logs.
- **Output:** model deteksi + dokumentasi.

System Call Tracing

Sample benign malware simulation

🔗 <https://github.com/forrest-orr/flare-floss>

Proyek 25 — Threat Intelligence Report (Malware Campaign)

- Analisis campaign malware tertentu:
 - IOC collection
 - TTP mapping
 - C2 infrastructure
 - propagation
- **Output:** laporan ala perusahaan TI (VirusTotal, Cyfirma, Kaspersky).

Threat Intelligence Report

Gunakan kampanye malware berikut (IOC only, safe):

- TA542 (Emotet)
- TA505 (Dridex)
- FIN6

Sumber IOC aman:

🔗 <https://otx.alienvault.com/>

🔗 <https://bazaar.abuse.ch/>

(IOC = hash, URL, domain → aman karena bukan executable)

CATATAN KEAMANAN (WAJIB UNTUK KELAS)

1. Selalu lakukan analisis di **VM isolated** (Windows 10 VM + snapshot).
2. Gunakan **Network set to Host-Only**, bukan NAT.
3. Jangan pernah mengeksekusi malware di komputer utama.
4. Gunakan **pwd: infected** untuk file dari TheZoo.
5. Gunakan **PCAP** dan **Memory Dump** untuk tugas yang tidak memerlukan malware executable.

Hasil Kegiatan UTS agar di Uploads pada :

https://drive.google.com/drive/folders/1_wPiw0Dk0kAXYPfvufkjr9WexVjQEX2b?usp=sharing



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